

$$f : \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$\begin{aligned}\max f(x) &= 5x_1 + 3x_2 - x_3 \\ x_1 + 2x_2 + x_3 &\leq 20 \\ 2x_1 + x_2 - x_3 &\geq 4 \\ x_2 + x_3 &= 6 \\ x_1 - x_2 &\leq 8 \\ x_1, x_2, x_3 &\geq 0\end{aligned}$$

Metodo delle due fasi

Fase I

$$\begin{aligned}\min f(x) &= a_1 + a_2 \\ x_1 + 2x_2 + x_3 + s_1 &= 20 \\ 2x_1 + x_2 - x_3 - s_2 + a_1 &= 4 \\ x_2 + x_3 + a_2 &= 6 \\ x_1 - x_2 + s_3 &= 8 \\ x_1, x_2, x_3, s_1, s_2, s_3, a_1, a_2 &\geq 0\end{aligned}$$

Calcolo dei costi

Forma Standard

$$\begin{aligned}\min f(x) &= -5x_1 - 3x_2 + x_3 \\ x_1 + 2x_2 + x_3 + s_1 &= 20 \\ 2x_1 + x_2 - x_3 - s_2 &= 4 \\ x_2 + x_3 &= 6 \\ x_1 - x_2 + s_3 &= 8 \\ x_1, x_2, x_3, s_1, s_2, s_3 &\geq 0\end{aligned}$$

$$c_i = c_N - c_B^\top \cdot B_N^{-1}$$

$$c_1 = 0 - (0; 1; 1; 0) \cdot \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = -2$$

$$c_2 = 0 - (0; 1; 1; 0) \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \\ -1 \end{pmatrix} = -2$$

$$c_3 = 0 - (0; 1; 1; 0) \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix} = 0$$

$$c_4 = 0 - (0; 1; 1; 0) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$c_5 = 0 - (0; 1; 1; 0) \cdot \begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \end{pmatrix} = 1$$

$$c_6 = 0 - (0; 1; 1; 0) \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$c_7 = 1 - (0; 1; 1; 0) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$c_8 = 1 - (0; 1; 1; 0) \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = 0$$

$$c_9 = 0 - (0; 1; 1; 0) \cdot \begin{pmatrix} 20 \\ 4 \\ 6 \\ 8 \end{pmatrix} = -10$$

Metodo del simplesso

	x_1	x_2	x_3	s_1	s_2	s_3	a_1	a_2	b
s_1	1	2	1	1	0	0	0	0	20
a_1	2	1	-1	0	-1	0	1	0	4
a_2	0	1	1	0	0	0	0	1	6
s_3	1	-1	0	0	0	1	0	0	8
	-2	-2	0	0	1	0	0	0	-10

Entra x_1 ed esce a_1 :

	x_1	x_2	x_3	s_1	s_2	s_3	a_1	a_2	b
s_1	1	2	1	1	0	0	0	0	20
a_1	2	1	-1	0	-1	0	1	0	4
a_2	0	1	1	0	0	0	0	1	6
s_3	1	-1	0	0	0	1	0	0	8
	-2	-2	0	0	1	0	0	0	-10

$R5 \leftarrow R5 + R2$

$$\begin{array}{cccccccc|c}
 -2 & -2 & 0 & 0 & 1 & 0 & 0 & 0 & -10 \\
 2 & 1 & -1 & 0 & -1 & 0 & 1 & 0 & 4 \\
 \hline
 0 & -1 & -1 & 0 & 0 & 0 & 1 & 0 & -6
 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	s_3	a_1	a_2	b
s_1	1	2	1	1	0	0	0	0	20
a_1	2	1	-1	0	-1	0	1	0	4
a_2	0	1	1	0	0	0	0	1	6
s_3	1	-1	0	0	0	1	0	0	8
	0	-1	-1	0	0	0	1	0	-6

$R2 \leftarrow R2 \cdot \frac{1}{2}$

	x_1	x_2	x_3	s_1	s_2	s_3	a_1	a_2	b
s_1	1	2	1	1	0	0	0	0	20
a_1	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	2
a_2	0	1	1	0	0	0	0	1	6
s_3	1	-1	0	0	0	1	0	0	8
	0	-1	-1	0	0	0	1	0	-6

$R4 \leftarrow R4 - R2$

$$\begin{array}{cccccccc|c}
 1 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 8 \\
 1 & \frac{1}{2} & -\frac{1}{2} & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & 2 \\
 \hline
 0 & -\frac{3}{2} & \frac{1}{2} & 0 & \frac{1}{2} & 1 & -\frac{1}{2} & 0 & 6
 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	s_3	a_1	a_2	b
s_1	1	2	1	1	0	0	0	0	20
a_1	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	2
a_2	0	1	1	0	0	0	0	1	6
s_3	0	$-\frac{3}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	6
	0	-1	-1	0	0	0	1	0	-6

$$R1 \leftarrow R1 - R2$$

$$\begin{array}{cccccccc|c} 1 & 2 & 1 & 1 & 0 & 0 & 0 & 0 & 20 \\ 1 & \frac{1}{2} & -\frac{1}{2} & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & 2 \\ \hline 0 & \frac{3}{2} & \frac{3}{2} & 1 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 18 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	s_3	a_1	a_2	b
s_1	0	$\frac{3}{2}$	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	18
x_1	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	2
a_2	0	1	1	0	0	0	0	1	6
s_3	0	$-\frac{3}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	6
	0	-1	-1	0	0	0	1	0	-6

Entra x_3 ed esce a_2 :

	x_1	x_2	x_3	s_1	s_2	s_3	a_1	a_2	b
s_1	0	$\frac{3}{2}$	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	18
x_1	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	2
a_2	0	1	1	0	0	0	0	1	6
s_3	0	$-\frac{3}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	6
	0	-1	-1	0	0	0	1	0	-6

$$R5 \leftarrow R5 + R3$$

$$\begin{array}{cccccccc|c} 0 & -1 & -1 & 0 & 0 & 0 & 1 & 0 & -6 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 6 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	s_3	a_1	a_2	b
s_1	0	$\frac{3}{2}$	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	18
x_1	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	2
a_2	0	1	1	0	0	0	0	1	6
s_3	0	$-\frac{3}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	6
	0	0	0	0	0	0	1	1	0

$$R1 \leftarrow R1 - R3 \cdot \frac{3}{2}$$

$$\begin{array}{cccccccc|c} 0 & \frac{3}{2} & \frac{3}{2} & 1 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 18 \\ 0 & \frac{3}{2} & \frac{3}{2} & 0 & 0 & 0 & 0 & \frac{3}{2} & 9 \\ \hline 0 & 0 & 0 & 1 & \frac{1}{2} & 0 & -\frac{1}{2} & -\frac{3}{2} & 9 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	s_3	a_1	a_2	b
s_1	0	0	0	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{3}{2}$	9
x_1	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	2
a_2	0	1	1	0	0	0	0	1	6
s_3	0	$-\frac{3}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	6
	0	0	0	0	0	0	1	1	0

$$R2 \leftarrow R2 + R3 \cdot \frac{1}{2}$$

$$\begin{array}{cccccccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & 2 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & \frac{1}{2} & 3 \\ \hline 1 & 1 & 0 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & \frac{1}{2} & 5 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	s_3	a_1	a_2	b
s_1	0	0	0	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{3}{2}$	9
x_1	1	1	0	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{2}$	5
a_2	0	1	1	0	0	0	0	1	6
s_3	0	$-\frac{3}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	6
	0	0	0	0	0	0	1	1	0

$$R4 \leftarrow R4 - R3 \cdot \frac{1}{2}$$

$$\begin{array}{cccccccc|c} 0 & -\frac{3}{2} & \frac{1}{2} & 0 & \frac{1}{2} & 1 & -\frac{1}{2} & 0 & 6 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & \frac{1}{2} & 3 \\ \hline 0 & -2 & 0 & 0 & \frac{1}{2} & 1 & -\frac{1}{2} & -\frac{1}{2} & 3 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	s_3	a_1	a_2	b
s_1	0	0	0	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{3}{2}$	9
x_1	1	1	0	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{2}$	5
x_3	0	1	1	0	0	0	0	1	6
s_3	0	-2	0	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	$-\frac{1}{2}$	3
	0	0	0	0	0	0	1	1	0

$z = 0$ si passa alla seconda fase.

Fase II

$$\min f(x) = -5x_1 - 3x_2 + x_3$$

$$s_1 + \frac{1}{2}s_2 - \frac{1}{2}a_1 - \frac{3}{2}a_2 = 9$$

$$x_1 + x_2 - \frac{1}{2}s_2 + \frac{1}{2}a_1 + \frac{1}{2}a_2 = 5$$

$$x_2 + x_3 + a_2 = 6$$

$$-2x_2 + \frac{1}{2}s_2 + s_3 - \frac{1}{2}a_1 - \frac{1}{2}a_2 = 3$$

$$x_1, x_2, x_3, s_1, s_2, s_3, a_1, a_2 \geq 0$$

Si eliminano le variabili artificiali introdotte nella Fase I

$$\min f(x) = -5x_1 - 3x_2 + x_3$$

$$s_1 + \frac{1}{2}s_2 = 9$$

$$x_1 + x_2 - \frac{1}{2}s_2 = 5$$

$$x_2 + x_3 = 6$$

$$-2x_2 + \frac{1}{2}s_2 + s_3 = 3$$

$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$

Calcolo dei costi della nuova tabella:

$$c_1 = -5 - (0; -5; 1; 0) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$c_2 = -3 - (0; -5; 1; 0) \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \\ -2 \end{pmatrix} = 1$$

$$c_3 = 1 - (0; -5; 1; 0) \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = 0$$

$$c_4 = 0 - (0; -5; 1; 0) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$c_5 = 0 - (0; -5; 1; 0) \cdot \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 0 \\ \frac{1}{2} \end{pmatrix} = -\frac{5}{2}$$

$$c_6 = 0 - (0; -5; 1; 0) \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$c_7 = 0 - (0; -5; 1; 0) \cdot \begin{pmatrix} 9 \\ 5 \\ 6 \\ 3 \end{pmatrix} = 19$$

Tabella risultante:

	x_1	x_2	x_3	s_1	s_2	s_3	b
s_1	0	0	0	1	$\frac{1}{2}$	0	9
x_1	1	1	0	0	$-\frac{1}{2}$	0	5
x_3	0	1	1	0	0	0	6
s_3	0	-2	0	0	$\frac{1}{2}$	1	3
	0	1	0	0	$-\frac{5}{2}$	0	19

Entra s_2 ed esce s_3 :

	x_1	x_2	x_3	s_1	s_2	s_3	b
s_1	0	0	0	1	$\frac{1}{2}$	0	9
x_1	1	1	0	0	$-\frac{1}{2}$	0	5
x_3	0	1	1	0	0	0	6
s_3	0	-2	0	0	$\frac{1}{2}$	1	3
	0	1	0	0	$-\frac{5}{2}$	0	19

$R1 \leftarrow R1 - R4$

$$\begin{array}{cccccc|c}
 0 & 0 & 0 & 1 & \frac{1}{2} & 0 & 9 \\
 0 & -2 & 0 & 0 & \frac{1}{2} & 1 & 3 \\
 \hline
 0 & 2 & 0 & 1 & 0 & -1 & 6
 \end{array}$$

$\downarrow$

	x_1	x_2	x_3	s_1	s_2	s_3	b
s_1	0	2	0	1	0	-1	6
x_1	1	1	0	0	$-\frac{1}{2}$	0	5
x_3	0	1	1	0	0	0	6
s_3	0	-2	0	0	$\frac{1}{2}$	1	3
	0	1	0	0	$-\frac{5}{2}$	0	19

$R2 \leftarrow R2 + R4$

$$\begin{array}{cccccc|c}
 1 & 1 & 0 & 0 & -\frac{1}{2} & 0 & 5 \\
 0 & -2 & 0 & 0 & \frac{1}{2} & 1 & 3 \\
 \hline
 1 & -1 & 0 & 0 & 0 & 1 & 8
 \end{array}$$

$\downarrow$

	x_1	x_2	x_3	s_1	s_2	s_3	b
s_1	0	2	0	1	0	-1	6
x_1	1	-1	0	0	0	1	8
x_3	0	1	1	0	0	0	6
s_3	0	-2	0	0	$\frac{1}{2}$	1	3
	0	1	0	0	$-\frac{5}{2}$	0	19

$R4 \leftarrow R4 \cdot 2$

	x_1	x_2	x_3	s_1	s_2	s_3	b
s_1	0	2	0	1	0	-1	6
x_1	1	-1	0	0	0	1	8
x_3	0	1	1	0	0	0	6
s_3	0	-4	0	0	1	2	6
	0	1	0	0	$-\frac{5}{2}$	0	19

$$R5 \leftarrow R5 + R4 \cdot \frac{5}{2}$$

$$\begin{array}{cccccc|c} 0 & 1 & 0 & 0 & -\frac{5}{2} & 0 & 19 \\ 0 & -10 & 0 & 0 & \frac{5}{2} & 5 & 15 \\ \hline 0 & -9 & 0 & 0 & 0 & 5 & 34 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	s_3	b
s_1	0	2	0	1	0	-1	6
x_1	1	-1	0	0	0	1	8
x_3	0	1	1	0	0	0	6
s_2	0	-4	0	0	1	2	6
	0	-9	0	0	0	5	34

Entra x_2 ed esce s_1 :

	x_1	x_2	x_3	s_1	s_2	s_3	b
s_1	0	2	0	1	0	-1	6
x_1	1	-1	0	0	0	1	8
x_3	0	1	1	0	0	0	6
s_2	0	-4	0	0	1	2	6
	0	-9	0	0	0	5	34

$$R1 \leftarrow R1 \cdot \frac{1}{2}$$

	x_1	x_2	x_3	s_1	s_2	s_3	b
s_1	0	1	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	3
x_1	1	-1	0	0	0	1	8
x_3	0	1	1	0	0	0	6
s_2	0	-4	0	0	1	2	6
	0	-9	0	0	0	5	34

$$R2 \leftarrow R2 + R1$$

$$\begin{array}{cccccc|c} 1 & -1 & 0 & 0 & 0 & 1 & 8 \\ 0 & 1 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 3 \\ \hline 1 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 11 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	s_3	b
s_1	0	1	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	3
x_1	1	0	0	$\frac{1}{2}$	0	$\frac{1}{2}$	11
x_3	0	1	1	0	0	0	6
s_2	0	-4	0	0	1	2	6
	0	-9	0	0	0	5	34

$$R3 \leftarrow R3 - R1$$

$$\begin{array}{cccccc|c} 0 & 1 & 1 & 0 & 0 & 0 & 6 \\ 0 & 1 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 3 \\ \hline 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} & 3 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	s_3	b
s_1	0	1	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	3
x_1	1	0	0	$\frac{1}{2}$	0	$\frac{1}{2}$	11
x_3	0	0	1	$-\frac{1}{2}$	0	$\frac{1}{2}$	3
s_2	0	-4	0	0	1	2	6
	0	-9	0	0	0	5	34

$$R4 \leftarrow R4 + R1 \cdot 4$$

$$\begin{array}{cccccc|c} 0 & -4 & 0 & 0 & 1 & 2 & 6 \\ 0 & 4 & 0 & 2 & 0 & -2 & 12 \\ \hline 0 & 0 & 0 & 2 & 1 & 0 & 18 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	s_3	b
s_1	0	1	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	3
x_1	1	0	0	$\frac{1}{2}$	0	$\frac{1}{2}$	11
x_3	0	0	1	$-\frac{1}{2}$	0	$\frac{1}{2}$	3
s_2	0	0	0	2	1	0	18
	0	-9	0	0	0	5	34

$$R5 \leftarrow R5 + R1 \cdot 9$$

$$\begin{array}{cccccc|c} 0 & -9 & 0 & 0 & 0 & 5 & 34 \\ 0 & 9 & 0 & \frac{9}{2} & 0 & -\frac{9}{2} & 27 \\ \hline 0 & 0 & 0 & \frac{9}{2} & 0 & \frac{1}{2} & 61 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	s_3	b
x_2	0	1	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	3
x_1	1	0	0	$\frac{1}{2}$	0	$\frac{1}{2}$	11
x_3	0	0	1	$-\frac{1}{2}$	0	$\frac{1}{2}$	3
s_2	0	0	0	2	1	0	18
	0	0	0	$\frac{9}{2}$	0	$\frac{1}{2}$	61

$$x_1 = 11$$

$$x_2 = 3$$

$$x_3 = 3$$

$$s_1 = 0$$

$$s_2 = 18$$

$$s_3 = 0$$

$$\min f(x) = -5x_1 - 3x_2 + x_3 \rightarrow -5(11) - 3(3) + (3) = -55 - 9 + 3 = -61$$

$$\max f(x) = 5x_1 + 3x_2 - x_3 \rightarrow \boxed{61}$$